BALOCHISTAN BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, QUETTA

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2024

MATHEMATICS

Past Paper 2024

Retyped faithfully from the original board paper — same questions, no changes.

PART - I

Note: Q.1 Choose the correct answer. (12x1=12)

Note: NOTE: Part-I is compulsory. Attempt eight (8) questions from Part-II, five (5) questions from Part-III and Two (2) questions from Part-IV. Only simple calculator and logarithm table is allowed.

Q1. [1 0; 0 3] is a ---------- matrix.

(A) Scalar
(B) Diagonal
(C) Rectangular
(D) Unit

Q2. In 2i(3 - i) the real part is '----------'.

(A) 1
(B) 2
(C) 3
(D) 6

Q3. In 1614, ---------- developed the concept of the logarithm.

(A) Al-Khawarizmi
(B) Briggs
(C) Napier
(D) Euler

Q4. The simplest form of √(1/4) is ----------.

(A) 2
(B) 3
(C) 1/2
(D) 2/3

Q5. Factors of $x^2 - 9y^2$ are ----------.

(A) $(x - 3y)^2$
(B) $(x + 3y)^2$
(C) $(3x - y)^2$
(D) $(x - 3y)(x + 3y)$

Q6. L.C.M of 4x, 8xy and $12x^2y^2$ is ----------.

(A) $24x^2y$
(B) $12x^2y$
(C) 24xy
(D) 12xy

Q7. The solution of $2z + 8 = 11$ is ----------.

(A) -3/2 (B) 3/2

(C) 2/3 (D) 3

Q8. (-1, -5) lies in ---------- quadrant.

(A) I (B) II

(C) III (D) IV

Q9. Two or more than two points which lie on the same straight line is called ---------- points.

(A) Non-collinear (B) Collinear

(C) Concurrent (D) line

Q10. The ---------- sides and angles of congruent triangles are congruent.

(A) Parallel (B) Opposite

(C) Corresponding (D) Unparallel

Q11. There are ---------- acute angles in right angled triangle.

(A) One (B) Two

(C) Three (D) Many

Q12. Pythagoras theorem was first derived by the Greek Mathematician Pythagoras ---------- years ago.

(A) 1500 (B) 2200

(C) 2300 (D) 2500

PART - II

Q2. (Marks 24) Q.2 Attempt any eight (08) questions. (8x3=24)

(i) Find the value of P when [2 -2][1; P] = [20].

(ii) Write (18)^1.3 in the radical form.

(iii) Find the value of x from $\log_8 64 = x$.

(iv) Simplify $(\sqrt{5} - \sqrt{8})^2$.

(v) Factorize $x^2 - 2x - 143$.

(vi) Find H.C.M of $x^3y^2z^4$, $x^2y^3z^2$, $x^2y^2z^3$ by factorization.

(vii) $6 \geq 4x \geq 3x - 1$ solve the inequalities for which $(x \in R)$

(viii) Find the midpoint of the following pair of points (-2, 1) and (6, 5).

(ix) Write in scientific notation the speed of sound 1236km/h after converting into cm/h.

(x) Define Isosceles triangle with diagram.

PART - III

> Note: Note: Attempt any Five (05) questions. (5x5=25)

Q3. (Marks 5) 4x + 3y = 6 use matrix inversion method to solve the simultaneous equations. x - 2y = 7

Q4. (Marks 5) With the help of logarithm, find the value of (32.6)^(2/3)

Q5. (Marks 5) If a + b + c = 9 and ab + bc + ca = 16 then find $a^2 + b^2 + c^2$.

Q6. (Marks 5) Factorize $(x^2 - 2x - 35)(x^2 - 2x - 3) - 144$.

Q7. (Marks 5) Find square root by division method of the following $4x^4 + 20x^3 + 41x^2 + 40x + 16$

Q8. (Marks 5) Solve the equation $\sqrt[3]{(2x + 3)} = \sqrt[3]{(3x - 2)}$ and check your solution.

Q9. (Marks 5) Show that points P(-3, -2), Q(2, -7) and R(-2, 5) are vertices of an isosceles triangle.

Q10. (Marks 5) Simplify 1/(2+3i) and write your answer in the form of a + ib.

PART - IV

> Note: Note: Attempt any two (2) questions. (7x2=14)

Q11. (Marks 7) Any point on bisector of an angle is equidistant from its arms.

Q12. (Marks 7) Triangles of equal bases and having the same altitudes are equal in area.

Q13. (Marks 7) Construct the ΔABC where mAB = 4.5cm, mBC = 3.9cm, and m∠B = 60°. Draw the bisectors of their angles.

Q14. (Marks 7) Prove that, in an equilateral triangle any two medians are congruent.